

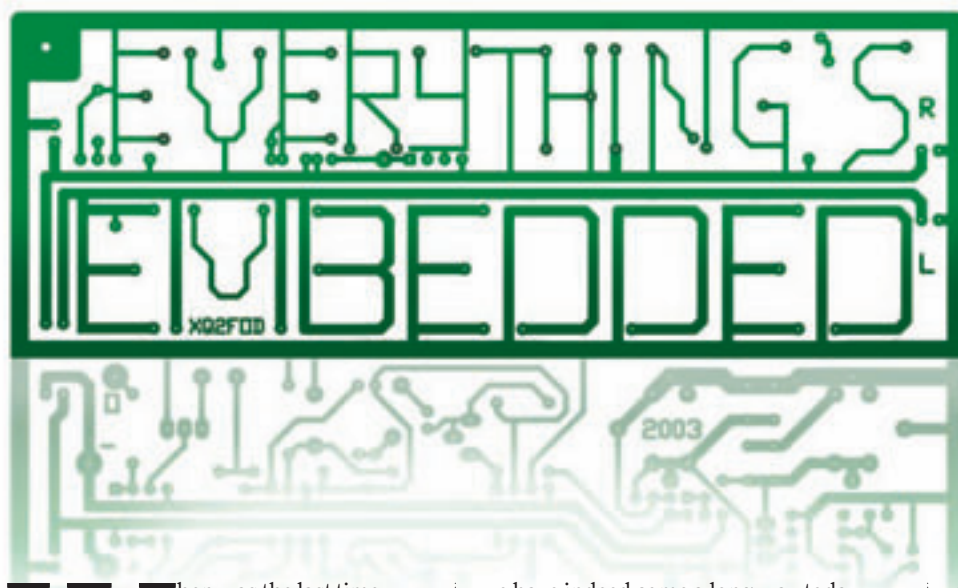
Penguin's Market

Linux OS has close to 90% of the supercomputer market in its grip



Elephantitis

IBM's ASCI white supercomputer weighs as much as 17 elephants and can do in one second what a calculator would take 10 million years to do



Totally unaware of the rapid development around us, we don't realise the importance of embedded systems in our lives

Jasnoor Gill
readersletters@thinkdigit.com

When was the last time you paid attention to the electronics behind the devices you interact with and use every single day of your lives? Over the last few decades, embedded technology has taken over our homes, being present today in almost every aspect of our lives. Most devices, from the humble electronic wristwatch to your car have one or even more embedded microprocessors in them.

These devices do all the work behind the scenes, silently and reliably. They work so well, in fact, that we rarely stop to even think of the mini-computer running inside the smallest and simplest (supposedly!) of devices. As a result, every house has dozens of embedded devices and microprocessors running in the background. For some, it may in fact even run into the hundreds.

Starting from the first true microprocessor invented in the early 1970s,

we have indeed come a long way today. Technology has kept advancing, resulting in increasingly powerful processors, while becoming even smaller at the same time. Beginning with some very specialised applications, embedded technology today is prevalent in every field imaginable. Today's automobiles are one of the best examples of the advances made in embedded technology. From the single microprocessor present in the 'trip-computer' of the 1978 Cadillac Seville, we have come a long way to today's high end Mercedes S-class and BMW 7-series models, boasting around 100 microprocessors. These control everything from the engine to the one present in each headlight! The electronics and related wiring is in fact a major part of the total cost of any car being manufactured today.

"How did we survive before the microprocessors took control of almost every machine we operate?" The answer lies in the demanding "we". We want a car which can do 0-100 kmph in 5 secs, yet it should give an economy of 15 kmpl. We want a phone which should have push mail, yet give 100 hours of talk time and give off minimum radiation. Such feats cannot be achieved by mere mechanical and electronic designs alone. The replacement of a carburettor with direct fuel injection would not have been possible without a microprocessor, because a carburettor is the best precision a mechanical design can achieve. Embedded designs help overcome such engineering and material limitations.

And this is the primary reason for their pervasiveness.

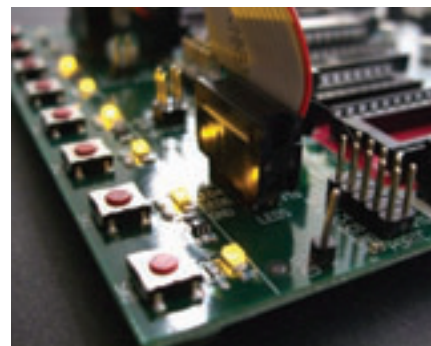
The second reason is our paucity of time. We want assistants, intelligent assistants. Our microwave oven should not overheat my food even if we're not around. Our washing machine should know how much detergent to use for our clothes, depending upon the type of clothes without us having to intervene. This "intervention-less" behaviour cannot be achieved without a microprocessor.

The third and most potent reason is connectivity. The "On Star" systems, installed in Ford vehicles in the US, can inform emergency services should the car meet with an accident. The BMW customer service can unlock and start your 7-series remotely if an emergency occurs. Closer to home, you can tell your Airtel set-top box to record your favourite program while you are stuck in a meeting. So in effect, you not only pay for the device which you have bought, you also pay for the connectivity and the associated ecosystem.

Microprocessors – everywhere and in everything

Here is some equipment which functions because of embedded technology:

- Multi-function electronic watches
- All computers/ATMs
- mobile phones, pagers and other communication devices
- Engine controllers, anti-lock brake (ABS) systems in cars
- Home appliances like microwave ovens, washing machines, televisions, air conditioners etc.
- All aircraft systems
- Safety systems like fire alarms, sprinklers



Embedded systems and controllers